Karim, Ehsan

From: Data science methodologies <saravanan.paramasivam@springernature.com>
Sent: December 13, 2024 4:22 AM
To: Karim, Ehsan
Subject: Decision on your submission to BMC Medical Research Methodology

[CAUTION: Non-UBC Email]

Ref: Submission ID 33ae6b4f-23c6-4525-911d-e0d6d645a507

Dear Dr Karim,

Your manuscript "Is There a Competitive Advantage to Using Multivariate Statistical or Machine Learning Methods Over the Bross Formula in the hdPS Framework for Bias and Variance Estimation?" has now been assessed. If there are any reviewer comments on your manuscript, you can find them at the end of this email.

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Handling Editor: Imran Ashraf

Comments:
Concerns have been raised by more than one reviewer concerning the technique of the study and the paper can not be accepted for publication.
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Regrettably, your manuscript has been rejected for publication in BMC Medical Research Methodology.

Thank you for the opportunity to review your work. I'm sorry that we cannot be more positive on this occasion and hope you will not be deterred from submitting future work to BMC Medical Research Methodology.

Kind regards,

Vrushali Mule
Editor
BMC Medical Research Methodology

Reviewer Comments:

Reviewer 1

Attachments:
• https://reviewer-feedback.springernature.com/download/attachment/1500c620-fb32-43bd-b4b7-0455437bdfe6

Reviewer 2

Attachments:
• https://reviewer-feedback.springernature.com/download/attachment/0e436acf-c6ba-4c93-991c-f1dc5a2c6b2e

Reviewer 3
The authors attempt to compare some common machine learning methods along with the Bross formula on a specific NHANES dataset. Because of a single focus on this dataset, which the authors admit of the limitations, the reviewer feels that the results do not sufficiently convey what the authors set out to do. The current paper exhibits a simple case study which may not have valid inference or reproducibility .

To improve on the paper, the authors should either present an extended case study of their own undertaking (consisting of study design, data collection, data clean up and analysis) or present multiple datasets along with more thorough simulations to make more sound inference and conclusions.

Reviewer 4
The authors address a pressing issue of how to estimate the propensity score (PS) in the high-dimensional setting with known confounders and selection of proxy variables. The paper is well-written and has some interesting results. The simulation design is specifically targeted towards real world scenarios. I commend the authors for looking the two aspects required for ideal estimation of the PS: we want to estimate the exposure effect in a way that is both unbiased and has the greatest precision. However, there are issues that remain to be addressed. Most critically is that the simulation is opaque in which variables are associated with exposure only, outcome only, true confounders , proxy variables, or complete noise variables. Was the level of confounding high or low? This makes the results impossible to interpret.

Additionally, it is disingenuous to include "deep learning" as a keyword when the extent of the paper on this topic is only one sentence in the discussion.

Major:
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1. The authors specifically focus on the proxy variable setting, where there are confounders that are not measured and we want to find proxy variables which are correlated (enough) with the unmeasured confounders such that we can adjust for confounding. The simulation design, to my understanding, does not seem to be designed this way. The inclusion of proxy variables are defined as variables with a RR < 0.8 or RR > 1.2 with the outcome. This seems to not require that the proxy variables have an association with the exposure -- so they may not even be true proxy variables. Please clarify if my understanding is wrong.
2. The plasmode simulation data generating model is not clear. Which variables were confounders and which variables were proxies? Were the investigator-specified covariates associated only with the outcome and not exposure? There are 94 proxy variables, but which of these were the demographic/behavior/health versus lab values versus ICD-10 codes?
3. Despite the other aspects to mimic real word data, all proxy variables in the simulation are binary. This is not realistic.
4. Why is the genetic algorithm even considered? I personally have not seen this used for variable selection anywhere. It shouldn't even be a contender, and the results are so bad it makes visualization of the other methods challenging.
5. Related to (1), what about the setting where not all confounders are known and it is assumed that there are additional confounders that are unknown and we want to select these, as opposed to proxy variables. There are many methods for this scenario in the high-dimensional setting that are included in the comparisons here, such as the outcome adaptive LASSO (https://pubmed.ncbi.nlm.nih.gov/28273693/).

Minor:
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1. Many of the ML methods used for PS estimation assume continuous predictors. Were categorical variables translated into dummy variables and excluded from the default centering and scaling that most ML implementations do?